Vehicle Health Analysis and Spare Parts E-Commerce

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Abstract — After a certain amount of time, vehicles require continual maintenance and repairs, and if they are not maintained at predetermined intervals, accidents may result. The project's goal is to provide the owner a general concept of the estimated insurance claim that he or she can receive in the event of specific faults and concerns, regardless of whether an insurance claim is actually made for the car or not. On-Board Diagnostic (OBD) and Engine Control Units (ECU), which inform the owner of the location of the fault, are just two of the many car health monitoring systems that are used to find and diagnose faults and problems. In this project, we've trained a machine learning algorithm that can forecast the expected insurance claim based on the faults that the owner has entered using Support Vector Regression, TFIDF, LSTM, and a dataset comprising automobile prices. This system's primary goal is to lessen the number of frauds that occur when the owner is unaware of the true cost. This project attempts to lessen it by assisting the owner in learning about the complete insurance coverage for their car that they are legally entitled to.

***Keywords* — On-Board Diagnostic, Engine Control Units, Support Vector Regression, TFIDF, LSTM**

I. INTRODUCTION

Many of today's vehicles include a wide range of systems and components that perform various operations while the vehicle is in use. Over time, repeated use of the vehicle may cause failure of individual systems or components. The failure of one component or system affects the overall performance of vehicle. For continuous better performance of vehicle, the best way is to carry out prescribed periodical inspection and repair in a prescribed way. It requires years of experience to inspect manually, thus the electronic vehicle health monitoring system is used. In order to get the maximum insurance sum of the vehicle that falls under the company's insurance claim policy, this is where vehicle insurance application comes which helps the owner to get maximum possible claim.

Insurance systems are widespread in today's world. They are being extensively deployed to check and verify the insurance sum of a particular entity. To name a few, insurance systems are used in healthcare, business, home, life and travel. Insurance systems uses the information of the various cost prices of the particular entities which they use along with the current issue in order to estimate the insurance claim.

Vehicle Insurance Application uses machine learning algorithms like Support vector regression, TFIDF and LSTM to predict an estimated price of the insurance sum based on the faults given by the owner. Thus, it helps in reducing scams like not paying the money, giving way less money than the actual amount, etc., and helps the owners to get their maximum possible claim.

II. CHALLENGES

- Dataset for this project was a big problem as many sites did not have a proper arranged data sets for analysis. Some had name and value while some had distance travelled but no other attribute. In this process there was a point where insufficient data was available for the machine to work, wherein prediction was not as expected.
- After a thorough search and time, we could find a dataset that could work with the analysis where it had horse power, milage, stokes and other import elements for a vehicle's engine
- When it comes to results, with the use of liner regression it was always possible for negative values which implies they are not worth insurance or the vehicle health was bad.
- These cases occurred frequently during the data analysis and prediction time which took a lot of time
- Even though the value is predicted, with more data on vehicle and previous insurance issued for those along with extra attributes can really help the machine provide more accurate value
- Also, use of other algorithms like SVM, logistic regression, Decision tree, KNN, K-mean can provide more insights on this prediction of value.

III. LITERATURE REVIEW

I. GAFFAR GULAB MOMIN

"A REVIEW ON VEHICLE HEALTH ANALYSIS MONITORING SYSTEM" [2022]

After a predetermined amount of time, vehicles require repair, and if they are not repaired at predetermined intervals, accidents may result. The project's goal is to continuously alert the owner to issues such as tire air pressure, exhaust gas quality, brake wear, oil leaks, excessive engine heating, fuel pipe blockages, etc. Tire pressure monitoring sensors are used to obtain real-time tire pressure readings. Gas sensors provide readings for carbon dioxide (CO2), nitrogen oxides (NOx), and other gases, and the quantity and quality of exhaust determine the engine's proper operating condition. Flow sensors are used to detect oil leaks and overflows, and temperature sensors to alert the owner when the engine is overheating. The LCD display unit, which is integrated with the Arduino microcontroller, shows all of these sensor data.

II. E. CHANDRASEKHAR

"VEHICLE HEALTH MONITORING SYSTEM" [2021]

The term "on-board diagnostics" in the automobile industry refers to a vehicle's capacity for self-diagnosis and reporting. The state of the various car subsystems can be accessed by the owner or a repair professional using OBD systems. Since its introduction in the first iterations of on-board car computers in the early 1980s, the amount of diagnostic data accessible via OBD has changed greatly. When a fault was discovered, older versions of OBD would only turn on a "idiot light" or malfunction indication light without giving any other details. Modern OBD implementations give real-time data via a standardized digital communications interface in addition to a standardized set of diagnostic trouble codes that enable quick identification and correction of vehicle malfunctions. OBD-I or OBD-II buses are typically included in cars to specify reports or diagnostics of the vehicle.

III. PARMESH K, RASHMI PRAFULLAKUMAR NERIYA, GANESAN K

"VEHICLE HEALTH MONITORING AND ANALYSIS" [2015]

Modern automobiles are more intelligent in terms of engine decisions, performance, fuel efficiency, security, and stability thanks to the integration of additional ECUs and sensors. Through the OBDII (on-board diagnostics) protocol, ECUs also provide helpful information in the form of diagnostic codes. These codes identify a particular problem in the vehicle's system and are used by service personnel to fix the problem. In order to monitor the condition of the vehicle's health in real-time, this article explores the possibility of OBDII codes from the engine ECU and computes the data on an Android device. The goal is to offer portable diagnosis solutions and include such systems into in-car entertainment systems like Apple CarPlay and Android Auto, which can be a crucial element for V2X and IOVs (Internet of vehicles).

IV. MING-SONG LIN, CUN-XIANG LIU

"RESEARCH OF THE SYSTEM OF REMOTING VEHICLE INSURANCE ESTIMATION" [2016]

The present model for estimating auto insurance has a lot of issues and takes too long. With the help of the remote vehicle insurance estimation system, the insurance provider will receive a picture of the vehicle loss situation in order to assess the damage promptly, make a quick decision, expedite processing of claims, reduce operating costs, and enhance the effectiveness of managing compensation. In contrast, it is clear that the remote vehicle insurance estimation system, which is a longer-range system than the vehicle insurance estimate model, is more suited to meet the demands of the market today. A significant promotion is required in the Chinese insurance industry.

V. UMA SHANKAR YADAV, RAVINDRA TRIPATHI, MANO ASHISH TRIPATHI

"REVIEW OF THE MOTOR VEHICLE INSURANCE INDUTRY IN INDIA" [2022]

The most popular and efficient way to safeguard personal property is through motor vehicle insurance, which has garnered a lot of interest. Understanding the present state of the Malaysian auto insurance market is essential. This setting serves as the foundation for this article's analysis of the Indian motor insurance sector's growth model and its connection to the country's economic growth. This article examines methods for achieving targeted product advertising and regionally-specific motor vehicle insurance (VMI). The goal of this paper is to build a solid foundation for promoting the growth of the Indian auto insurance market and its connection to the country's vehicle manufacturing. The article examines the impact of Malaysia's VMI policy on the nation's economy and motor vehicle production in comparison to other developed nations. In addition, the essay emphasizes how COVID-19 has affected the VMI sector.

VI. JACOB MEHLMAN, JIAHUI WU, HANSHEN JING JINGYI QI

"PREDICTION OF CUSTOMER'S INTERESTS IN VEHICLE INSURANCE" [2022]

The insurance provider that previously offered health insurance to its customers must now determine whether those same policyholders (customers) from the previous year will be interested in the provider's offering of automobile insurance. In order to forecast clients' propensity to purchase vehicle insurance, we built statistical models utilizing bagging, AdaBoost, random forest, XGBoost, and convolutional neural networks based on demographic data, vehicle scenario, and insurance policy information. To compare the models, we provide ROC curves, confusion matrices, and accuracy scores. We then select the CNN model and offer advice to the insurance provider.

IV. DATASET

The suggested model was tested using the gathered data, which included cost price of the vehicle and the general information like the mileage, cylinders, fuel type model, etc. A total of 51887 information about the vehicles it contains. To carry out the experiment, 85%of the dataset is utilized for training, 15% for validation and the remaining 5% for testing. Table shows total number of different types of vehicles used.

Type	Number of vehicles
Bike (fig.1)	32649
Car (fig.2)	17687
Truck (fig.3)	1551
Total	**51887**

(fig.1)

(fig.2)

(fig.3)

V. METHODOLOGY

Vehicle analysis is a complex task due the number of challenges in data collection and training appropriately on the processed data. This task can be tackled by using regression models like Support Vector Regression.

Starting with the collection of data and preparing/preprocessing dataset on **vehicles** and the **Three most prevalent types** that are:

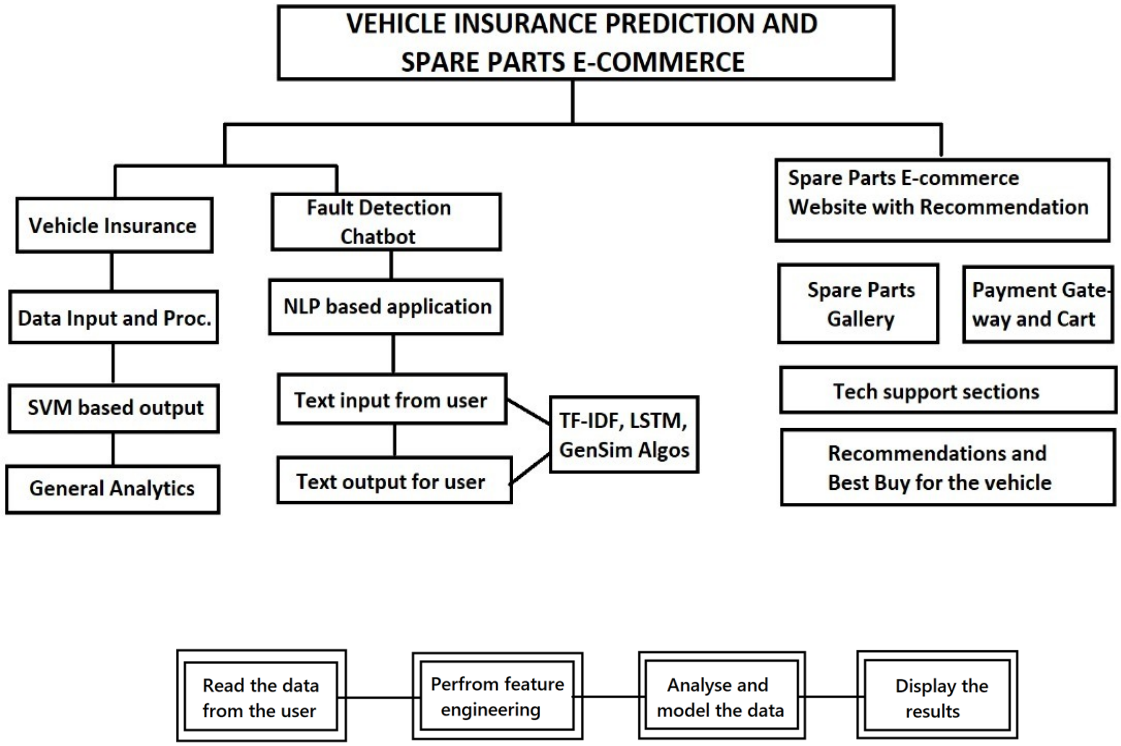
1. Bike [fig.1]
2. Car [fig.2]
3. Truck [fig.3]

Data analysis utilising various methods can be used to predict the value for insurance of the car. With the data provided, including mileage, journey distance, horse power, and many other specifics, we are employing the fundamental process known as Linear Regression for the predictor. In order to determine a value based on the condition of the vehicle, these variables are plotted against one another.

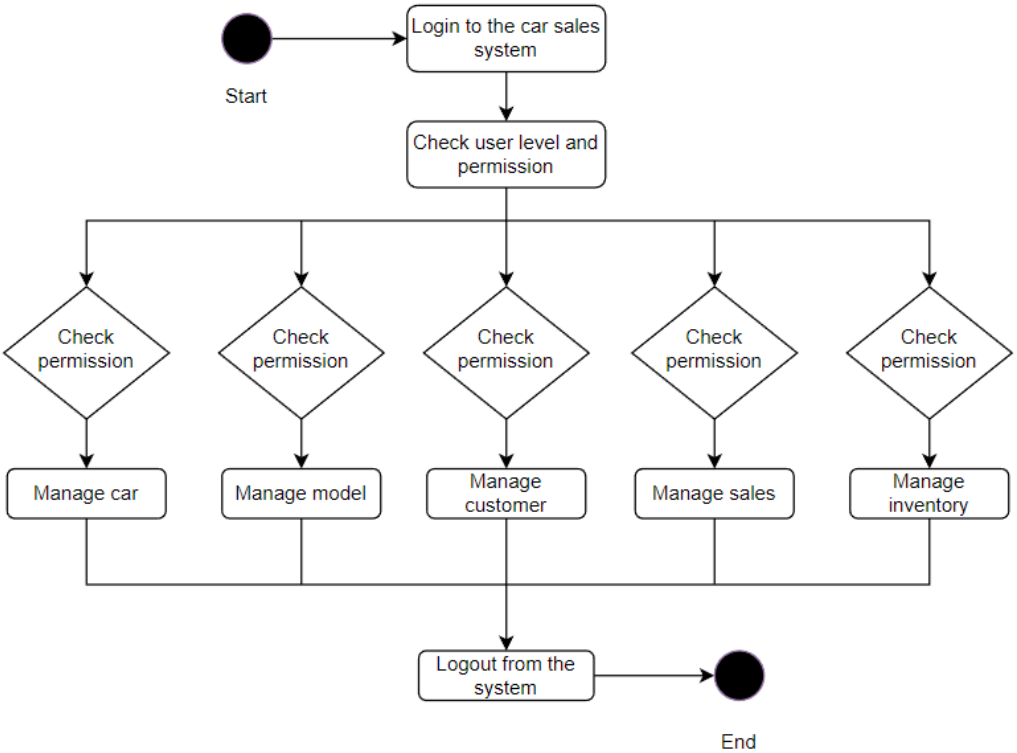
This programme is designed to forecast a potential cost for a vehicle's insurance. This analyses data from over 20,000 data samples to calculate the potential insurance pay out. This programme is designed to estimate the potential cost of a used motorcycle in India.

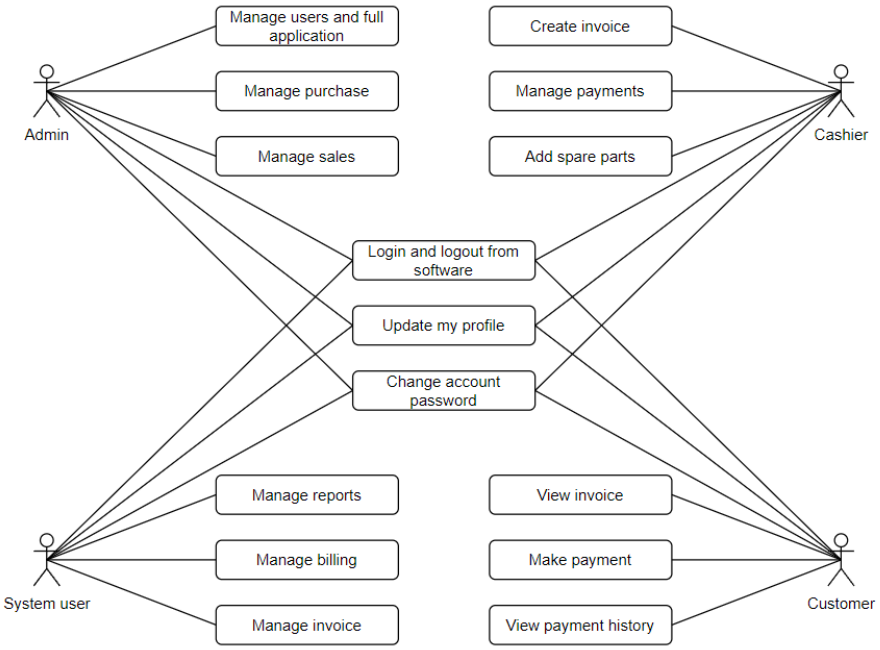
To determine the optimum price, the programme uses preloaded motorcycles data that includes 30k+ data samples of used bikes and their sell prices, 23 companies, and several attributes. The value is estimated after careful data analysis, even if it may not be entirely correct. It might be enhanced by introducing a new method and feeding the machine new data to train it.

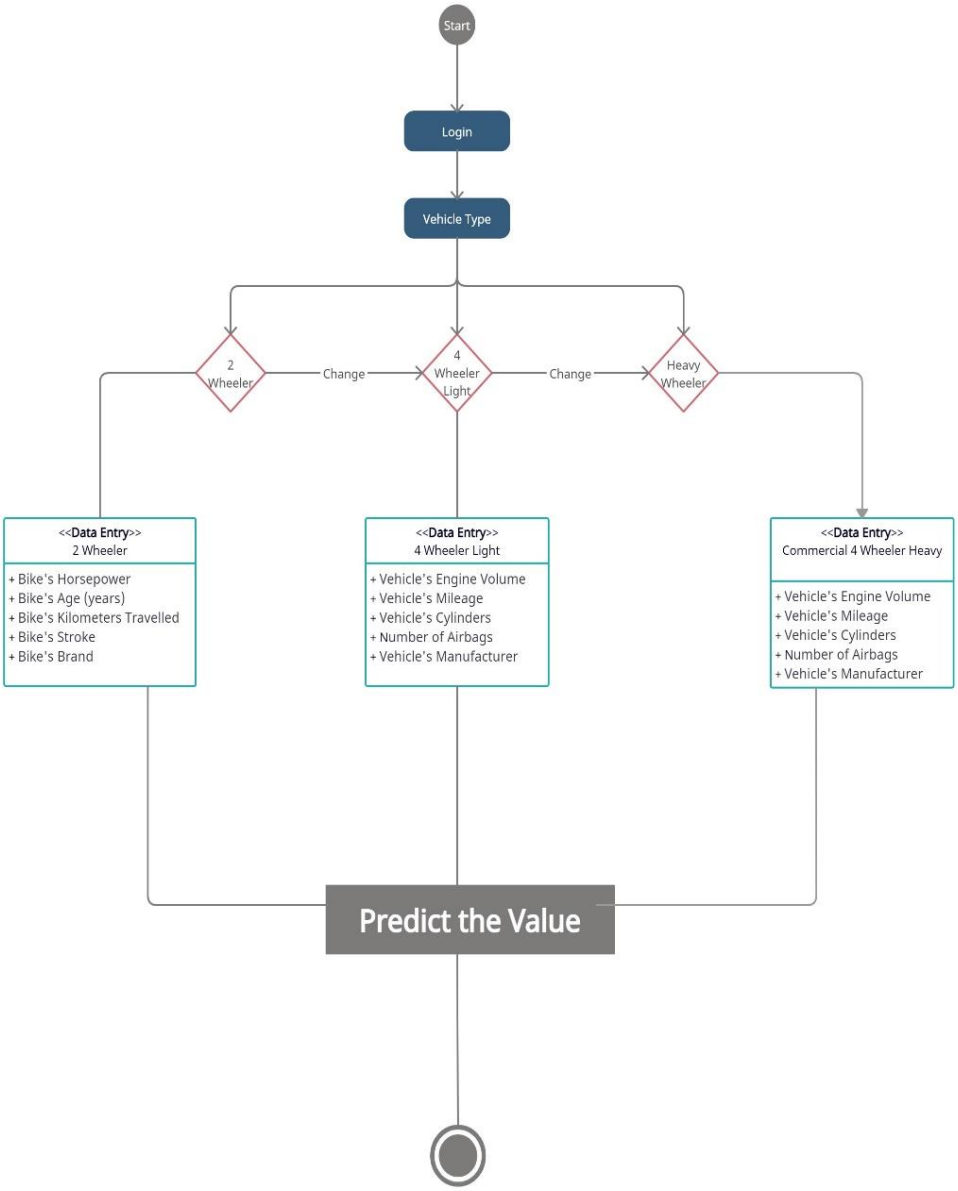
Architecture Diagram:



UML Diagrams:




Manage users and full application
Manage purchase
Manage sales
Admin
Create invoice
Manage payments
Add spare parts
Cashier
Login and logout from software
Update my profile
Change account password
Manage reports
Manage billing
Manage invoice
System user
View invoice
Make payment
View payment history
Customer


Start
Login
Vehicle Type
2 Wheeler
Change
4 Wheeler Light
Change
Heavy Wheeler
<<Data Entry>>
2 Wheeler
+ Bike's Horsepower
+ Bike's Age (years)
+ Bike's Kilometers Travelled
+ Bike's Stroke
+ Bike's Brand
<<Data Entry>>
4 Wheeler Light
+ Vehicle's Engine Volume
+ Vehicle's Mileage
+ Vehicle's Cylinders
+ Number of Airbags
+ Vehicle's Manufacturer
<<Data Entry>>
Commercial 4 Wheeler Heavy
+ Vehicle's Engine Volume
+ Vehicle's Mileage
+ Vehicle's Cylinders
+ Number of Airbags
+ Vehicle's Manufacturer
Predict the Value

Algorithms Used:

Support Vector Regression:

SVMs or Support Vector Machines are one of the most popular and widely used algorithm for dealing with classification problems in machine learning. This algorithm acknowledges the presence of non-linearity in the data and provides a proficient prediction model. The SVM regression algorithm is referred to as Support Vector Regression or SVR. In machine learning, Support Vector Machines are supervised learning models with associated learning algorithms that analyze data used for classification and regression analysis. In Support Vector Regression, the straight line that is required to fit the data is referred to as hyperplane. The objective of a support vector machine algorithm is to find a hyperplane in an n-dimensional space that distinctly classifies the data points. The data points on either side of the hyperplane that are closest to the hyperplane are called Support Vectors. These influence the position and orientation of the hyperplane and thus help build the SVM.
This algorithm allows scanning through multiple parameters and understand the nature of insurance premium depending on varied factors of cars and bikes.

TFIDF and LSTM:

These have been used in the NLP part of the application where an user can input his or her problem statement and the auto bot will be replying in human understandable sentences and also suggest a best buy from the website.
TFIDF stands for term frequency and inverse document frequency. This model finds out the most stressed words and the problem topic and helps to find a solution accordingly. LSTM stands for Long short-term memory. LSTM model is used to find the sequentiality in the text. This means that in a sentence, the LSTM model will consider the words in continuation as a human reads a sentence and not pickup individual words and analyze the meaning. This is particularly helpful while answering the user in sentence and this can be very useful for unforeseen use cases as well.
The third part is a simple ecommerce website with admin panel facilities made using PHP and MySQL. The purpose of this is to sell the spare parts and also provide free replacement facilities for the customer.

Modules Used:

<u>Module 1</u>: Data Integration

Data from a excel is integrated to the machine for data analysis. This data is then pre-processed using LabelEncoder(). Data of vehicles with attributes like engine volume, Mileage, Distance travelled, Horse Power, Stokes, Airbags, Bike company are integrated in to representing as a graph.

<u>Module 2</u>: ML Algorithm

Using python libraries, we integrate ML algorithms like SVM, Liner regression and many others to analysis the data. In this project we used only liner but other algorithms are also possible in which we can predict the values but making a slight change in the source code.

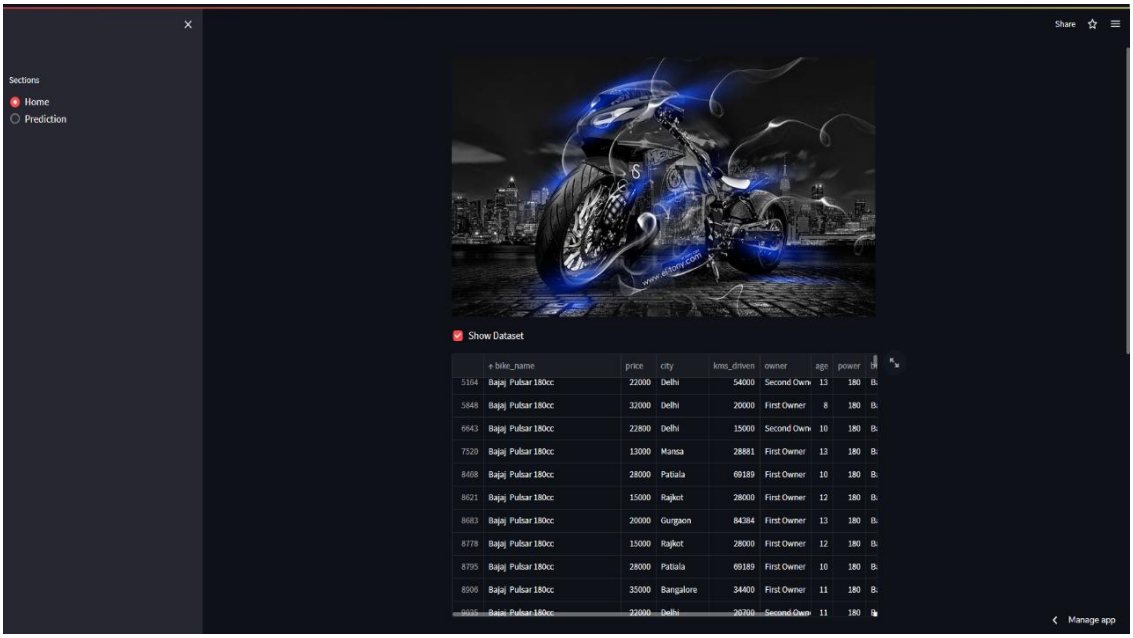
<u>Module 3</u>: Value prediction and visualization

In this module we have data visualization when in graphs are being introduced to show relation between the attributes but plotting the graphs using matplotlib library. And using ML we predict the value to be paid as well. This prediction is selected by analysis of the data in the CSV files we integrated.

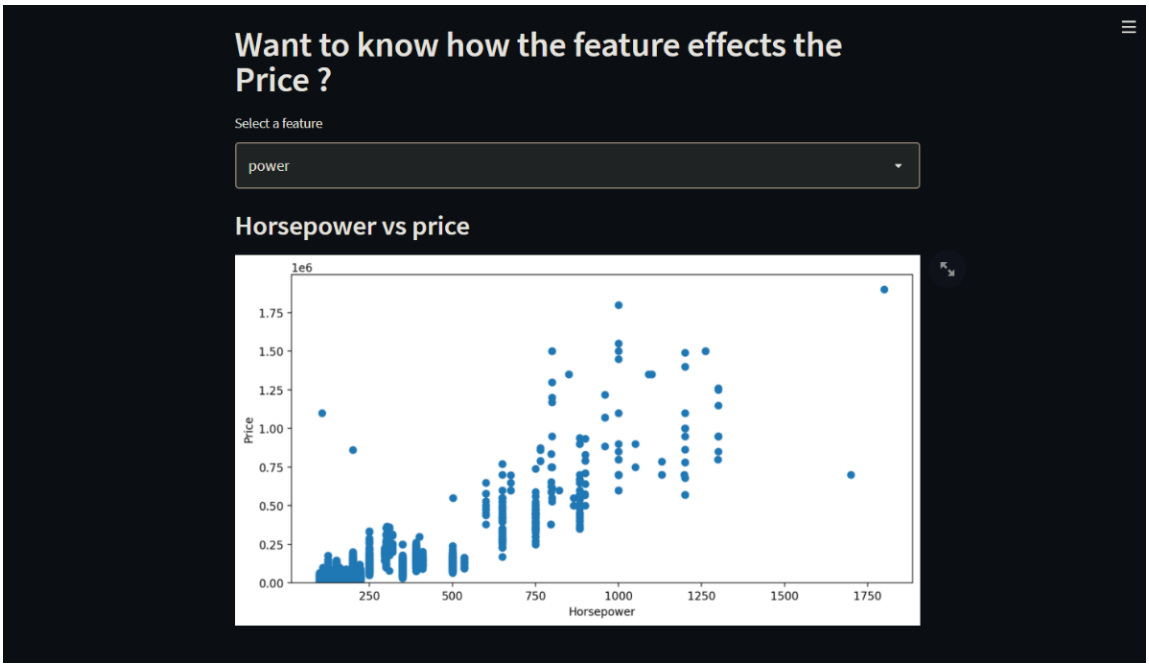
Python Libraries Used:

- **Streamlit** is an open-source app framework in Python language. It helps us create web apps for data science and machine learning in a short time.
- **Pandas** is an open-source Python package that is most widely used for data science/data analysis and machine learning tasks.
- **NumPy** is a Python library used for working with arrays. It also has functions for working in domain of linear algebra, Fourier transform, and matrices.
- **Matplotlib** is a comprehensive library for creating static, animated, and interactive visualizations in Python.
- **Pyplot** is an API (Application Programming Interface) for Python's matplotlib that effectively makes matplotlib a viable open-source alternative to MATLAB
- **Sklearn** contains a lot of efficient tools for machine learning and statistical modeling including classification, regression, clustering and dimensionality reduction.

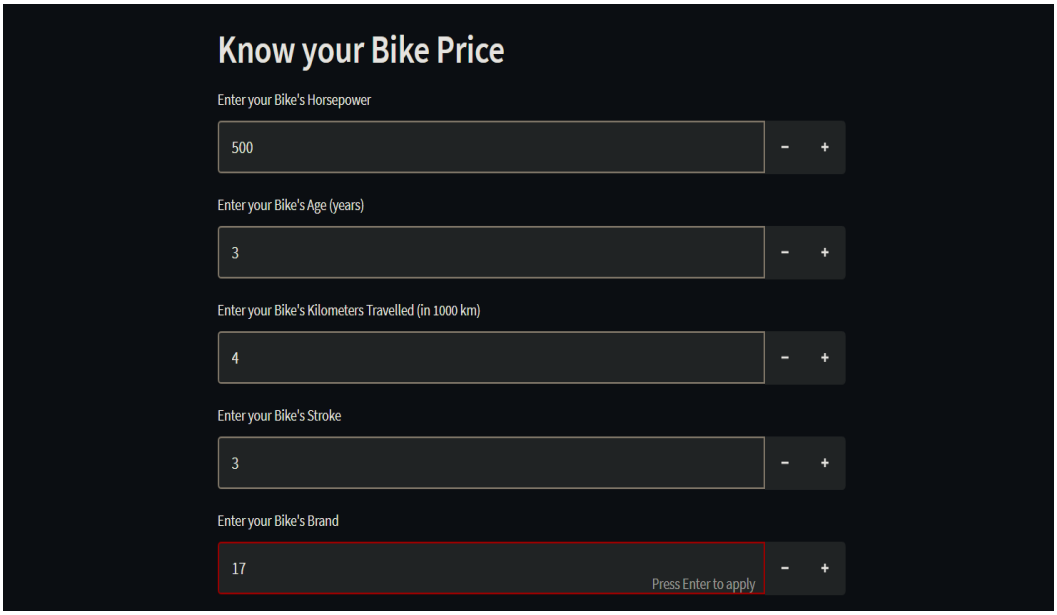
VI *RESULT*



Home page and Dataset

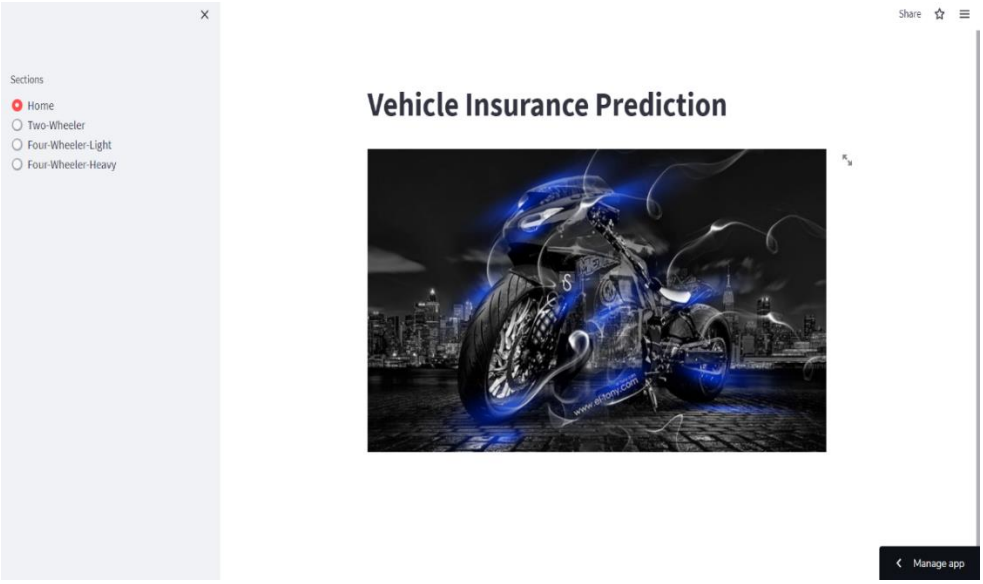


Data Visualization of dataset

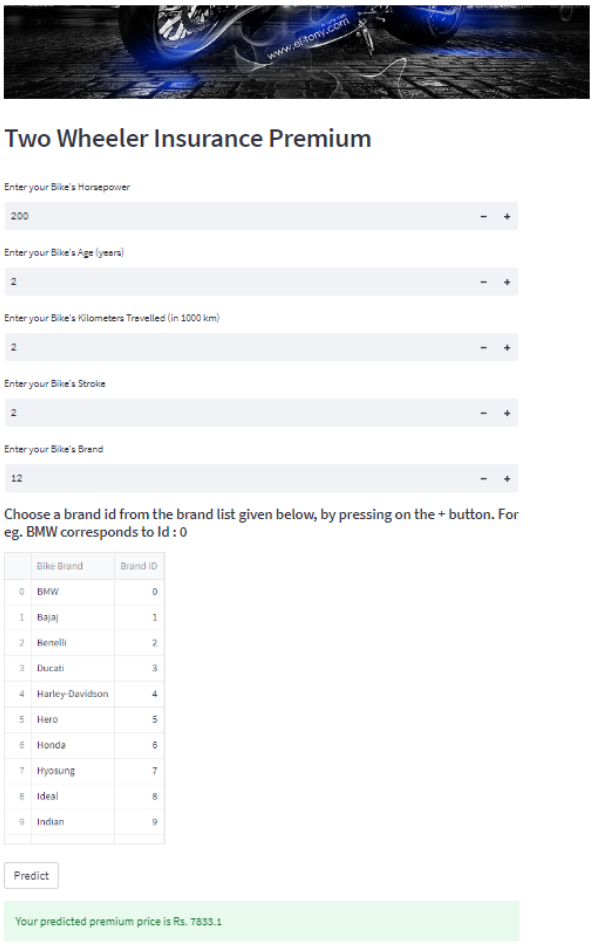


Data Entry for analysis

Price prediction



By using ML algo we analysed the data and integrated it to UI design as shown above. In right, image shows the prediction of premium for the bike data entered.



<u>Output-demo of The Project</u>

VI. CONCLUSION

The vehicle monitoring system offers accurate information on all fundamental factors influencing a vehicle's performance. Due to the early detection of damage made possible by this information, the performance of the vehicle will improve. As a result, it gives the owner accurate information so that during car servicing, the service centers cannot present incorrect information and demand additional payment.

The vehicle insurance app aids the owner in obtaining the most insurance payout possible in the event that the vehicle develops flaws or malfunctions as a result of normal wear and tear. According to the company's insurance policy, the owner can obtain the greatest claim possible, and that too without deceit or fraud. This method of auto insurance aids in eradicating such offensive behaviors.

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